

the b+g high-speed standard. As far as LAN is concerned, only the Acer, Dell and Fujitsu LBS6240 had Gigabit Ethernet, while the rest had the slower 10/100. A 56 Kbps modem was standard on all laptops. A PCMCIA Type II slot was available on all the machines except the Dell Latitude X1 and Samsung Q30 Plus.

Multimedia

Needless to mention, a microphone-in and headphone-out were available on all the laptops. A D-Sub connector was also available on all except the Fujitsu LBS6240, probably due to its larger 13.3-inch display which, along with that of the Sony VIAO, was the largest in this category. The Fujitsu LBP7010 was the only laptop with an S-Video Out and a TV-out.

Multimedia and memory card reading capability was available at least in some capacity on most of laptops.

The Dell supported only two formats: SD and CF. Acer supported SD, MS, MS Pro and MMC, whereas the Fujitsu LBP7010 supported the SD, MS, MS Pro and CF formats. Sony VIAO was the only laptop that supported the Memory Stick Duo standard.

Other features

The Fujitsu laptops boasted the highest-capacity hard drives—80 GB. MSI's laptops had the lowest-capacity hard drives of 40 GB.

All laptops had 4200 rpm hard drives, except for the Sony VIAO, which had a faster, 5400 rpm, 60 GB hard drive.

Most of the laptops had integrated graphics solutions based on either the Intel Extreme Graphics 2 or the Intel GMA 900. Though these are OK for office work, they are just plain bad as far as even moderate gaming is concerned.

The MSI MegaBook S270 had ATi Xpress 200P integrated graphics but this was throttled down due to the lesser amount of RAM; it therefore did not do well in the gaming tests. The Sony VIAO shone here with the nVidia GeForce Go 6200 32MB PCI-Express graphics. It blazed ahead of the competition in the gaming tests.

An integrated optical drive, too, was featured in most of the laptops. The only laptops with an external drive were the Acer, Dell and Samsung's Note PC Q30 Plus.

Although the Fujitsu LBP7010 was the smallest laptop, the optical drive was integrated into the unit. While most came with a combo drive, the Samsung Q30 Plus had a DVD+RW drive, and

How We Tested

Here's a description of the tests and benchmarks we used to compare the laptops.

Features

Features such as the size of the LCD, the number of I/O ports, hard disk size, type of optical disk drive, etc. were noted and rated. Extra features, if present, were also noted. Weightages were assigned to the noted features according to their importance.

Usability

The usability of a laptop is evaluated on the basis of how easy it is to handle in day-to-day life. Importance was given to weight and dimensions, as well as the ergonomics of the keyboard, touchpad, etc. We played music files to rate the speakers on a scale of five.

Package contents

Did the vendor provide an operating system, recovery CD and driver CDs with the laptop? We also looked at the extra software provided, as well as bundled accessories—power adapter, carry case, etc.

Performance

To gauge performance, we ran a battery of tests that evaluated each individual sub-system. We used all the following benchmarking suites:

PCMark 2004 v1.30 is a system-wide benchmark that tests the complete system as well as individual components such as the processor, memory, hard drive and graphics subsystem. It returns an aggregated score as well as individual scores by running applications used in daily work such as file encryption applications, virus scanning applications and so on.

SiSoft Sandra 2005 Professional SR2 is another system-wide benchmark suite. We used it to evaluate the performance of the CPU, memory and hard disk. The scores for the individual sub-systems were noted.

Ziff Davis Business Winstone 2004 tests the system by performing a scripted run of applications that are used on a daily basis, such as MS Word, PowerPoint, Excel, etc., as well as

applications such as Norton AntiVirus and WinZip. The overall system performance is evaluated, and a unified score is returned at the end of the run. The higher this score, the better is the system's performance.

Some of the other tests we did were:

Video encoding: A 100 MB VOB file was encoded using Dr. DivX 1.0.6 to convert it to AVI, and the encoding time was noted.

Gaming benchmarks: We wanted to test both OpenGL as well as Direct3D graphics capabilities. *Call of Duty* tests the OpenGL subsystem. The game was run at 800 x 600 and at 1024 x 768, and the average fps was noted. *FarCry* was the new game we included. It is a highly scalable game and uses Direct3D API. The game was run at 640 x 480 and at 1024 x 768, and the average fps was noted.

LCD screen display tests: DisplayMate Video Edition was used to gauge the sharpness and level shift of the LCD screens. We also used the LCD Pixel Persistence test from the PassMark Monitor Test suite; here, white moving blocks were observed for the tearing effect, and the screens rated accordingly.

Digit Battery Meter: In order to stress the battery to the utmost, we ran a VOB file—a DVD rip—after recharging the battery to capacity until the battery ran out. The battery time reveals whether the notebook will last a full movie—pretty real world. During this test, the screensaver as well as the power management features were disabled. Also, the volume level was maximised, the brightness was left at half, and the Wi-Fi was turned off.

Wi-Fi Test: To evaluate the data throughput over Wi-Fi, we used NetCPS—a utility that pumps data to check the TCP/IP connection and returns a score that reflects the throughput. We also copied 100 MB of assorted data and noted the time taken for the transfer.

Flaunt Value: Since this was a comparison test of 'lifestyle' laptops, we decided to include a new parameter—'Flaunt Value'. Since this is a completely subjective rating, we decided that a plebiscite was in order. We asked each voter to rate the laptops on a scale of 10 for design and looks, form factor, build quality, features and specs, package contents and brand value. An average of the ratings for each parameter was calculated and the average of all parameters for a particular laptop was recorded as its Flaunt Value.